2021 HW2

Dual Thresholds, a Rotator, and Ratioed Logic

EE3032 / Introduction to VLSI Design / Jin-Fu Li

Provenance. The original assignment and the supplied answers by Lin Yan-Yu were read first. The results below were independently derived, with the useful reference choices retained and improvements identified. Source details and hashes are in the year-level `SOURCES.md`.

1. High-V_t selection on the NAND branch

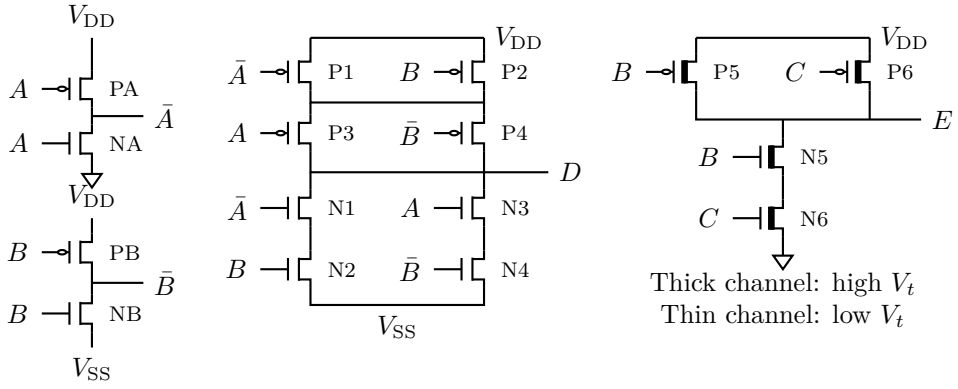
From the original gate symbols,

$$D = \bar{A} \oplus B = AB + \bar{A}\bar{B}, \qquad E = \overline{BC}.$$

With the given one-unit low-V_t gate model, the path $A \to \textit{inverter} \to XOR \to D$ is critical at two units. The direct $B \to XOR \to D$ and the two NAND paths each take one unit. Keep the inverter and XOR realization low-V_t; replace the *entire* NAND driving E by high-V_t devices.

Path	Low-V_t delay	New delay	Slack before change
$A \to D$	2	2	0
$B \to D$	1	1	1
$B, C \to E$	1	$d_{\mathrm{HVT,NAND}}$	1

Necessary timing condition: $d_{\mathrm{HVT,NAND}} \leq 2$ units. The prompt supplies no high-V_t delay or threshold values, so a numerical leakage reduction or unconditional timing guarantee cannot be inferred. Size the high-V_t NAND if necessary to satisfy this condition. Changing transistors inside the shared critical XOR without a library timing model is not justified by the direct-B slack alone. Below is a fully complementary static-CMOS realization. Complement-driver outputs are labeled rather than routed across the schematic; connect each identically named gate to its driver. All pMOS bodies connect to VDD and all nMOS bodies to VSS. The B complement generator belongs to the XOR-cell implementation, within the given abstract cell-delay model.



Reference comparison. The supplied answer selects the NAND branch for high V_t correctly. We retain that choice and state the missing timing assumption. The netlist `dual_vt_logic.sp` exposes symbolic LVT/HVT models; CIC018 TT does not supply this dual-V_t PDK. Eight exhaustive Boolean tests pass. The runnable `dual_vt_boolean_testbench.sp` additionally checks all eight states using explicitly illustrative Level-1 models; it does not verify dual-V_t process leakage or timing.

2. Active-low one-hot cyclic rotator

Only the four listed control words are specified. Let $r = 0, 1, 2, 3$ when $c_{3-r} = 0$ and the other controls are one. Then

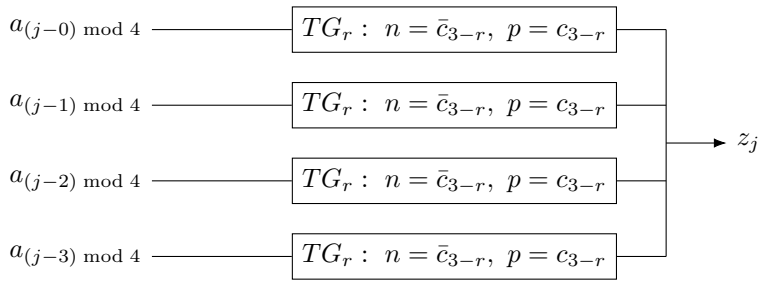
$$\boxed{z_j = a_{(j-r) \bmod 4}}, \quad j = 0, 1, 2, 3.$$

Under that legal-control assumption, the direct sum of selected paths is

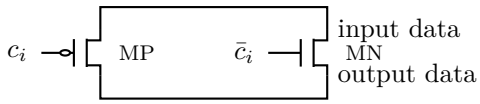
$$z_j = \bar{c}_3 a_j + \bar{c}_2 a_{(j-1) \bmod 4} + \bar{c}_1 a_{(j-2) \bmod 4} + \bar{c}_0 a_{(j-3) \bmod 4}.$$

$c_3c_2c_1c_0$	z_3	z_2	z_1	z_0	r
0111	a_3	a_2	a_1	a_0	0
1011	a_2	a_1	a_0	a_3	1
1101	a_1	a_0	a_3	a_2	2
1110	a_0	a_3	a_2	a_1	3

Generate the four $\bar{c}_i$ values once with CMOS inverters. For each output z_j, connect four transmission gates in parallel. Path r connects $a_{(j-r) \bmod 4}$ to z_j, using nMOS gate $\bar{c}_{3-r}$ and pMOS gate c_{3-r}. Exactly one path is on for every specified control word.



Transmission-gate transistor expansion: each box contains two MOSFETs with the same data terminals, shown below. The small pMOS gate bubble denotes active-low conduction.

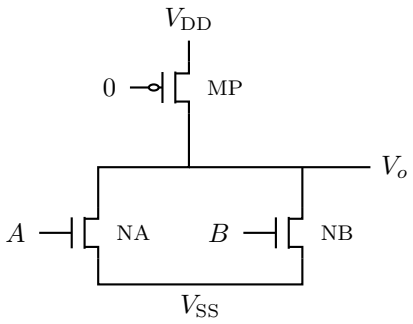


There are $4 \times 4 \times 2 = 32$ pass transistors plus eight inverter transistors: **40 transistors**. A data path crosses one TG and passes full rail voltages. Multiple zero controls would connect distinct input sources together, and all-one controls disconnect the outputs; these patterns are outside the specified domain. **Reference comparison.** The supplied schematic decodes each four-bit control word with four serial TGs per input path. It is valid for the listed controls, but has more devices and four pass stages. The direct one-hot implementation above is smaller. `verify_logic.py` and actual CIC018 HSPICE both pass all $16 \times 4 = 64$ cases (256 output-bit checks), with 1 fF on each output.

3. Pseudo-NMOS NOR and its size constraint

The pull-up pMOS has a grounded gate and is always on. The two parallel nMOS devices form the pull-down OR network. Hence

$$\boxed{V_o = \overline{A + B}}.$$



Let s_n and s_p multiply the common reference aspect ratio W/L. By the specified constant-resistance model,

$$R'_n = R_n/s_n, \qquad R'_p = R_p/s_p = 2R_n/s_p.$$

The worst low level is when exactly one nMOS is on; two on devices halve the pull-down resistance. Thus

$$V_{OL,\max} = V_{\mathrm{DD}} \frac{R'_n}{R'_p + R'_n} \le 0.1 V_{\mathrm{DD}} \iff R'_p \ge 9R'_n \iff \boxed{s_n \ge \tfrac{9}{2} s_p.}$$

Both nMOS must satisfy this bound independently. With equal lengths choose $(W/L)_n = 4.5(W/L)_p$. Equivalently choose $s_n = 9, s_p = 2$, as in the supplied answer. The ratio is ≥ 4.5, not a required absolute transistor width. For $R_n = 1\,\mathrm{k\Omega}$, $s_n = 9, s_p = 2$, and $V_{\mathrm{DD}} = 1.8$ V:

A	B	Constant-R V_o (V)	Actual CIC018 TT V_o (V)
0	0	1.800000	1.800000
0	1	0.180000	0.030165
1	0	0.180000	0.030122
1	1	0.094737	0.014822

Physical check. `pseudo_nor.sp` chooses $W_p = 1\,\mu$m, $W_n = 4.5\,\mu$m, and $L = 0.18\,\mu$m. Real HSPICE at TT, 25°C, and 1.8 V meets the 0.18 V low-level bound. BSIM is not the prompt's constant-R model, so the smaller observed values are supplementary process-specific results, not an alternative derivation. Ratioed logic dissipates static current when pulled low. **Reference comparison.** The supplied Boolean NOR and $(s_n, s_p) = (9, 2)$ selection are correct and retained. This solution states the full inequality and verifies both the specified resistance model and the local process model. **Reproduce.** Run `python verify_logic.py`, then `python reproduce.py -model D:/cic018.l -work PATH`. The simulator defaults to the installed HSPICE path and accepts `-hspice`. Outputs are in `results/`; raw logs stay in the specified work directory. No process-model text is distributed.